CE 1201

Engineering Mechanics Master Study Guide

Set-wise Class Notes · Formula · Theory/Proof · Question Bank Analysis · Explained Solutions · Backlog Preparation

Khulna University of Engineering & Technology (KUET)
Department of Building Engineering & Construction Management
Full Marks: 210 · Time: 3 hours

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Prepared from the supplied course notes, lecture slides, regular question banks, backlog paper and solution reference. 2020 is treated as an exception to the normal set pattern as instructed.

How to use this guide

Read set → revise formulas → write proofs → solve high-frequency patterns → attempt paper.

Set rule: A regular question paper contains four full questions in each part. Each full question is treated as one “set”.
2020: 2020 is treated as an exception to the normal set pattern, following the instructor note supplied by the course owner.

Part A Class Note

Set A1 · Centroid, Center of Gravity & Center of Mass

The geometric center family: centroid of line/area/volume, C.G., C.M., symmetry, standard shapes and composite areas.

Priority: Must Do · Source: 1 - Centroid - by 2223007; Shuvo Dip Datta lecture slides; Hamid + Shuvodip class note

Centroid — মূল ধারণা

Centroid হলো একটি uniform-density geometrical shape-এর geometric center. Area problem-এ এটি এমন একটি point যেখানে পুরো area-কে concentrated ধরে first moment সমান রাখা যায়.

$$\bar{x} = \frac{\int x\, dA}{\int dA}, \quad \bar{y} = \frac{\int y\, dA}{\int dA}$$

মনে রাখুন: Centroid = geometry-এর কেন্দ্র; weight দরকার নেই। Composite area হলে “Area × distance” table বানানো সবচেয়ে নিরাপদ।

QB: Definition is note-tagged for 2015–2022 and the topic continues in 2023–2025 papers.

Center of Gravity vs Center of Mass

Center of Gravity (C.G.) হলো যে point দিয়ে body-এর resultant gravitational weight কাজ করে। **Center of Mass (C.M.)** mass distribution-এর unique point.

$$\bar{x} = \frac{\int x\, dm}{m}, \quad \bar{y} = \frac{\int y\, dm}{m}, \quad \bar{z} = \frac{\int z\, dm}{m}$$

Uniform and parallel gravitational field-এ **C.G. = C.M.**. Uniform density body-তে mass density cancel করলে C.M. geometrical centroid-এর সাথে coincide করে.

QB: 2023 asks distinction/applications; 2024/2025 ask conditions for coincidence.

Principle of Symmetry

If a plane figure has a line of symmetry, its centroid lies on that line. If it has two intersecting symmetry axes, their intersection gives the centroid.

Shortcut: Symmetry দেখলেই একটি coordinate সরাসরি fix করুন; unnecessary integration করবেন না।

QB: Axis of symmetry is note-tagged in 2017, 2018 and 2020.

Composite Area Method

1. Shape-কে rectangle/triangle/circle/sector ইত্যাদিতে ভাগ করুন.
2. Reference X-Y axes fix করুন.
3. প্রতি অংশের A, x, y বের করুন.
4. Hole/cut-out-এর area negative নিন.
5. $\bar{x} = \Sigma Ax/\Sigma A$ এবং $\bar{y} = \Sigma Ay/\Sigma A$.

Exam table: Part | A | x | Ax | y | Ay. এই table ভুল sign কমায়।

QB: Composite centroid numerical is one of the most persistent patterns across regular papers.

Triangle Centroid Derivation

Base *b*, height *h*. A horizontal strip at height y has width b(h−y)/h.

$$dA = \frac{b(h-y)}{h} dy$$

Then $\bar{y} = \int y\, dA/A = h/3$ from the base. Similarly, for a right triangle with legs on axes, $\bar{x}=b/3$ and $\bar{y}=h/3$.

QB: Triangle derivation is note-tagged in 2016, 2018 and 2021.

Arc and Sector of a Circle

For a circular **arc** of radius r subtending 2β:

$$\bar{x} = \frac{r \sin \beta}{\beta}$$

For a circular **sector** of radius r subtending 2β:

$$\bar{x} = \frac{2r \sin \beta}{3\beta}$$

β must be in radians for these formulas.

QB: Sector centroid appears repeatedly, including 2023 and 2025.

Parabolic Boundary — high-value advanced problem

For the first-quadrant area bounded by $y^2=4x$ and $x^2=4y$, intersections are (0,0) and (4,4). By symmetry $\bar{x}=\bar{y}$.

$$A = \int_0^4 (2\sqrt{y} - y^2/4)dy = 16/3$$

$$\bar{x} = \bar{y} = 9/5 = 1.8 \text{ in}$$

এই problem-এ symmetry আগে ধরলে calculation অর্ধেক হয়ে যায়।

QB: Prepared centroid note labels this as a 2022 question; 2024 asks the same family of parabolic-centroid problem.

Set A2 · Moment of Inertia of Area

Rectangular and polar second moment of area, radius of gyration, standard-shape formulas and parallel-axis theorem.

Priority: Must Do · Source: 03 - Slide - Moment of Inertia of Area; 201-SD~1; Section A class notes

Second Moment of Area

Area moment of inertia describes how an area is distributed about an axis and directly influences beam stress/deflection calculations.

$$I_x = \int y^2 dA, \quad I_y = \int x^2 dA$$

Unit is length^4: mm^4, in^4, m^4.

QB: Definition and applications recur throughout the question bank.

Polar Moment of Inertia

For z-axis perpendicular to the area:

$$J_O = \int r^2 dA = I_x + I_y$$

কারণ $r^2=x^2+y^2$; তাই perpendicular-axis theorem খুব সহজে prove করা যায়।

QB: J = Ix + Iy proof is a standard theory question.

Radius of Gyration

$$k_x = \sqrt{(I_x/A)}, \quad k_y = \sqrt{(I_y/A)}, \quad k_O=\sqrt{(J/A)}$$

It is the distance at which the whole area could be imagined concentrated to give the same MOI.

QB: Often paired with polar MOI numericals.

Parallel-Axis / Transfer Theorem

$$I = \bar{I} + Ad^2$$

$\bar{I}$ is MOI about a parallel centroidal axis, d is the perpendicular distance between axes.

Rule: আগে centroid বের করুন, তারপর d. Hole হলে both area contribution and its transferred MOI are subtracted.

QB: A core theorem and calculation tool in nearly every composite MOI problem.

Standard Formula — Rectangle & Triangle

Rectangle

$I_{x,G}=bh^3/12$

$I_{y,G}=hb^3/12$

Triangle

$I_{x,base}=bh^3/12$

$I_{x,G}=bh^3/36$

QB: Triangle derivation and composite-section use are recurrent.

Solved L-Section Pattern

For a 2×8 in vertical rectangle plus a 7×2 in top rectangle:

$$\bar{x} = 2.167 \text{ in}, \quad \bar{y} = 6.333 \text{ in}$$

$$I_{x,G} = 276.67 \text{ in}^4, \quad I_{y,G} = 109.17 \text{ in}^4$$

> এই pattern-এ সবচেয়ে common ভুল: top rectangle-এর y-coordinate 8+1=9 in না নেওয়া।

QB: Matches the composite-transfer workflow used in the supplied class note.

Set A3 · Product of Inertia, Mass MOI & Pappus–Guldinus

The advanced inertia set: product of inertia/principal values, inertia of masses and surface/volume of revolution.

Priority: Very High · Source: 3.1 Product of Inertia; 3.2 Moment of Inertia of Masses; Pappus-Guldinus slides

Product of Inertia

$$P_{xy} = \int xy\, dA$$

Unlike Ix or Iy, Pxy may be positive, negative or zero. If x or y is an axis of symmetry, Pxy=0.

$$P_{xy} = \bar{P}_{xy} + A\, d_x d_y$$

QB: The supplied note tags a Z-section product-inertia problem for 2017–2022.

Principal / Maximum-Minimum MOI

$$I_{avg}=(I_x+I_y)/2$$

$$R=\sqrt{\{[(I_x-I_y)/2]^2+P_{xy}{}^2\}}$$

$$I_{max,min}=I_{avg}\pm R$$

k=√(I/A).

QB: 2023 explicitly asks maximum/minimum centroidal MOI for an angle section.

Verified Z-Section Example

For the 8×5×1 in Z-section used in the prepared note:

$$P_{xy}=70\text{ in}^4,\ I_x=141.34\text{ in}^4,\ I_y=61.33\text{ in}^4$$

$$I_{max}=181.96\text{ in}^4,\ I_{min}=20.71\text{ in}^4$$

$$k_{max}=3.372\text{ in},\ k_{min}\approx 1.138\text{ in}$$

QB: A very high-yield repeated pattern.

Moment of Inertia of Masses

$$I = \int r^2 dm, \quad I = \bar{I} + md^2$$

US customary problems require mass in **slug**: m=W/g.

> Area MOI-তে A; Mass MOI-তে m — parallel-axis formula-র এই পার্থক্য ভুলবেন না।

QB: Mass inertia is used in disk/shaft, frustum, mallet and impact/dynamics problems.

Standard Mass MOI

Slender rod, center

$I=mL^2/12$

Solid sphere

$I=2mr^2/5$

Solid cylinder, axis

$I=mr^2/2$

Solid disk

$I=mr^2/2$

QB: 2024 asks cylinder; 2025 asks sphere; 2023 includes disk/shaft.

Mallet Example

Supplied solution: 3 ft handle weighs 3.14 lb; cylindrical head weighs 16.16 lb. Using transfer theorem:

$$I_{y,total} \approx 4.00\ \text{slug}\cdot\text{ft}^2$$

$$k_y = \sqrt{(I/m)} \approx 2.59\ \text{ft}$$

QB: The same mallet family appears in 2025.

Pappus–Guldinus Theorem I — Surface

Surface generated by revolving a plane curve of length L about an external axis:

$$S = 2\pi\bar{y}L$$

$\bar{y}$ is the perpendicular distance of the curve centroid from the axis of revolution.

QB: Cone-paint and surface-of-revolution questions recur.

Pappus–Guldinus Theorem II — Volume

Volume generated by revolving a plane area A:

$$V = 2\pi\bar{y}A$$

For a torus formed by rotating a circle radius a whose center is b from the axis: $V=2\pi^2a^2b$.

QB: 2018/2023/2025 use pulley rim/solid of revolution variants.

Set A4 · Dynamics, Work–Energy & Impulse–Momentum

Classical dynamics using force–mass acceleration, work–energy, impulse–momentum and projectile/plane motion.

Priority: Must Do · Source: 4.1 Dynamics; 4.2 Impulse Momentum; 4.3 Work, Power and Energy; Shuvo Dip class notes

Classical Dynamics Map

Kinematics describes geometry of motion. **Kinetics** connects motion with its causes.

Kinematics → displacement, velocity, acceleration
Kinetics → F=ma, Work–Energy, Impulse–Momentum

QB: 2023 directly asks "What is classical dynamics?"

Force–Mass Acceleration

$$\Sigma F = ma$$

On an incline, resolve forces parallel/perpendicular to the plane before applying F=ma.

> FBD → choose positive direction → friction direction → ΣF=ma. এই sequence follow করলে sign ভুল কমে।

QB: Prepared note contains several block-and-pulley acceleration problems.

Work and Kinetic Energy

$$W_{net} = \Delta KE = \tfrac{1}{2}m(v_2^2 - v_1^2)$$

Work of a constant force is F s cosθ. Friction work is negative when opposing motion.

QB: Work–energy proof and incline numericals are highly repeated.

Verified Incline Work Example

Backlog-style data: W=200 lb, α=35°, Q=125 lb, β=25°, μ=1/3, s=12 ft.

$$N = W\cos\alpha - Q\sin\beta = 111.00 \text{ lb}$$

$$F_f = \mu N = 37.00 \text{ lb}$$

$$R_{along} = Q\cos\beta - W\sin\alpha - F_f = -38.43 \text{ lb}$$

$$W_{net} = Rs \approx -461.13 \text{ ft·lb}$$

Negative net work ⇒ kinetic energy decreases.

QB: This numerical family is directly visible in the 2023 backlog paper.

Linear Impulse–Momentum

$$\int_{t1}^{t2} F\,dt = m\,v_2 - m\,v_1$$

For constant resultant R: $R\Delta t=\Delta(mv)$.

> Impulse = force × time; Momentum = mass × velocity. Unit consistency is essential.

QB: A standard proof in 2016/2018/2023/2024 and related papers.

Angular Impulse–Momentum

$$\int M_O\,dt = I_O(\omega_2-\omega_1)$$

Used for disk/shaft deceleration and rotating-body problems.

QB: 2023 disk + shaft problem is a direct application.

Projectile / Plane Motion Shortcut

$$x=v\cos\theta\cdot t,\quad y=v\sin\theta\cdot t-\tfrac{1}{2}gt^2$$

For the recurring motorcycle problem (70 mph, 30°, 20 ft ditch, launch point 10 ft above ground), the supplied-note method gives:

$$h_{max}\approx 20.73\text{ ft}$$

> প্রথমে horizontal motion দিয়ে time; তারপর vertical equation দিয়ে height.

QB: Motorcycle problem is note-tagged in 2015, 2018 and 2021.

Part B Class Note

Set B1 · Force System, Equilibrium & Free-Body Diagrams

Definitions, support reactions, resultant, two-force members, transmissibility, FBD technique, wheel-over-block and boom/derrick patterns.

Priority: Must Do · Source: 10-EQU~1; Kabbo Sir force-system note; Arif Sir class note

Free-Body Diagram

An FBD isolates a body/member and shows all external forces: weight, applied load, support reactions, tension/compression and friction.

1. Identify the member to isolate.
2. Remove surroundings and replace contacts/supports by reactions.
3. Add gravity unless explicitly neglected.
4. For two-force members, forces are equal, opposite and collinear.

QB: FBD definition is note-tagged for 2015,16,17,18,21,22 and continues in 2023–2025.

Transmissibility of Force

For a rigid body, moving a force anywhere along the same line of action without changing magnitude or direction does not alter the external equilibrium/motion effect.

Conditions: same line of action, same magnitude, same direction.

QB: One of the most repeated short definitions.

Equilibrium

$$\Sigma F_x=0,\quad \Sigma F_y=0,\quad \Sigma M=0$$

For a 3D rigid body, corresponding z-force and x/y/z moment equations are added.

QB: Central to every statics numerical.

Support Reactions

Roller
1 reaction normal to surface
Pin/Hinge
R_x, R_y
Fixed
R_x, R_y, M

> Support দেখেই unknown count লিখুন; তারপর equation count মিলান।

QB: Frequently embedded inside FBD/beam questions.

Resultant of Coplanar Concurrent Forces

$$R=\sqrt{[(\Sigma F_x)^2+(\Sigma F_y)^2]}$$

$$\theta=\tan^{-1}[(\Sigma F_y)/(\Sigma F_x)]$$

Always correct θ according to the quadrant of ΣFx, ΣFy.

QB: Appears in force-system questions and class-note practice.

Wheel Over a Block — Least Force

Recurring note problem: W=1000 lb, radius=3 ft, block geometry giving α=56.81°.

$$F = \frac{W\cos 56.81°}{\sin(\theta+56.81°)}$$

Least F occurs when denominator is maximum: sin(θ+56.81°)=1.

$$\theta=33.19°,\quad F_{min}=547.42\text{ lb}$$

QB: Note-tagged for 2015,16,17,21; backlog 2023 contains the same family.

Two-Force Tension / Compression

Tension forces act away from the member; compression forces act toward the member. Rope/cable is tension-only; a weightless boom is commonly treated as a compression/two-force member.

QB: Used in derrick/boom and truss questions.

Set B2 · Shear Force, Bending Moment & Pulley

Equivalent distributed loads, reactions, shear/bending at a cut and pulley tension/reaction chains.

Priority: Must Do · Source: 2.1 SFBM; 2.2 Pulley; Kabbo Sir notes

Shear Force & Bending Moment Definitions

Shear force at a section = algebraic sum of external forces parallel to the section on either side.

Bending moment at a section = algebraic sum of moments of all external forces about the section on either side.

QB: Both definitions are note-tagged for 2016,17,18,19,21,22.

Load Resultants

UDL w over L
W=wL at L/2
Triangular UVL
W=½wL at L/3 from heavy end
Trapezoid
split into rectangle + triangle
Point moment
adds to BM, not shear

QB: Correct load replacement is the key to almost every beam numerical.

SFBM Exam Workflow

1. Replace UDL/UVL by equivalent point loads.
2. Draw support reactions.
3. Find reactions from ΣM=0 and ΣV=0.
4. Cut at the requested section.
5. Compute V and M from either left or right; both must agree.

> Cut-এর দুই side-এ একই answer না এলে sign/lever arm check করুন।

QB: The supplied note labels a core beam pattern for 2018–2021.

Verified Beam Example

In the prepared SFBM example with UDL, 30-kip point load, 20-kip-ft couple and triangular load:

$$V_B = 109\ \text{kip}, \quad V_E = 36\ \text{kip}$$

$$\text{At section (i): } V = -21\ \text{kip}, \quad M = -48\ \text{kip·ft}$$

The same result is obtained from either side of the cut.

QB: Excellent template for 2018/2019/2021-style beam questions.

Pulley Core Rule

For an ideal rope over frictionless, weightless pulleys, tension is constant along a continuous rope. A movable pulley supported by two equal rope segments has 2T upward.

> Pulley chain-এ “একটা object কতগুলো tension segment দিয়ে supported?” সেটাই আগে count করুন।

QB: Pulley is repeated in 2016,17,18,20,22 note tags and 2023–2025 papers.

Pulley + Beam Example

Prepared Q1: ideal pulley system connected to a beam carrying 700 lb and 2 kip/ft UDL.

$$\Sigma M_A = 0 \Rightarrow W = 21{,}500 \text{ lb}$$

Prepared Q2 family gives W=5,400 lb; another gives W=4,800 lb. The difference comes from the number of supporting rope segments.

QB: Use the exact rope routing in the figure; never memorize W/4 or W/8 without counting.

Set B3 · Truss & Plane Motion

Truss assumptions, method of joints/sections, tension-compression identification, and the plane-motion/projectile crossover.

Priority: Very High · Source: 3.1 Truss by 2223012; 3.2 Plane Motion; class notes

Truss Definition & Assumptions

A truss is a framework of straight members arranged mainly in triangles.

- Joints are idealized as frictionless pins.
- Loads act at joints.
- Members are two-force members and carry axial tension/compression.
- Bending of members is neglected in the ideal truss model.

QB: Definition/assumptions recur with numerical member-force calculation.

Method of Joints

Find support reactions first. Start at a joint with at most two unknown member forces. Assume unknowns tension (arrows away); a negative result means compression.

$$\Sigma F_x=0, \quad \Sigma F_y=0$$

QB: Best when many/all member forces are required.

Method of Sections

Pass a cut through no more than three unknown members, isolate one side and use equilibrium.

$$\Sigma M=0, \quad \Sigma F_x=0, \quad \Sigma F_y=0$$

> যে member force দ্রুত চান, অন্য দুই cut-member-এর intersection point-এ moment নিন।

QB: Common in Howe roof truss questions.

Verified Truss Pattern

For the supplied sloping-truss example, reactions are first found from whole-truss equilibrium. A section through the target members gives:

$$F_{GJ} \approx 9.928 \text{ kips (Compression)}$$

The prepared solution then proceeds member-by-member with section/joint equations.

QB: Related to 2017/2022-tagged truss examples and later regular papers.

Plane / Rectilinear Motion

Plane motion: every point of a body moves in planes parallel to a fixed plane. Rectilinear translation is straight-line motion where all points have the same displacement along parallel lines.

QB: Often appears as a short definition combined with a mechanics numerical.

Projectile Crossover

The motorcycle jump belongs to plane motion/particle projectile analysis and can be solved from $x = v\cos\theta \cdot t$ and $y = v\sin\theta \cdot t - \frac{1}{2}gt^2$.

QB: Recurring in historical papers and the dynamics set; cross-linked here intentionally.

Set B4 · Friction, Wedge, Belt Friction & Flexible Cord

Limiting friction, angle of repose, sliding/tipping, wedge mechanics, belt friction and cable/parabolic/catenary equations.

Priority: Must Do · Source: 3.3-4.1-4.2 Friction; 4.3 Flexible Cord; regular/backlog papers

Friction Basics

$$F \le \mu N, \quad F_{lim} = \mu N$$

Static friction self-adjusts up to the limiting value. Kinetic friction acts during sliding and opposes relative motion.

QB: Laws of friction and related numericals are persistent.

Angle of Friction & Angle of Repose

Angle of friction φ: angle between resultant reaction R and normal N at limiting friction.

$$\tan\varphi = \mu$$

Angle of repose α: inclination at which a body is just about to slide under its own weight. At impending slide, $\alpha=\varphi$ and $\tan\alpha=\mu$.

QB: Difference between the two appears repeatedly, including backlog 2023 and 2024/2025.

Sliding vs Tipping Decision

1. Calculate force required for impending sliding using $F=\mu N$.
2. Calculate force required for impending tipping by taking moment about the likely tipping edge.
3. The smaller required force determines what happens first.

> **Never decide from μ alone.** Geometry/height of force application controls tipping.

QB: A classic repeated question across the regular and backlog papers.

Belt Friction Proof Result

For a small belt element dθ at impending motion, resolving forces and using $dF=\mu dN$ gives $dT/T=\mu d\theta$.

$$\ln(T_1/T_2)=\mu\theta$$

$$T_1=T_2e^{\mu\theta}$$

θ must be in radians.

QB: A direct proof question in 2015, backlog 2023 and 2024.

Wedge Problems

Draw a separate FBD for wedge and each contacting block. For smooth contact reaction is normal; with friction, resultant reaction is inclined by $\varphi=\tan^{-1}\mu$ opposite impending motion.

> Wedge-এ contact force direction ঠিক করতে impending motion আগে লিখুন।

QB: Wedge appears in 2016 and recent regular/backlog papers.

Parabolic Flexible Chord

When load is uniform per horizontal length, a cable approximates a parabola.

$$y = \frac{wx^2}{2H}$$

$$H = \frac{wL^2}{8d}$$

$$T = \sqrt{[H^2+(wx)^2]}$$

At support x=L/2, tension is maximum.

QB: Parabolic cable equations are common in 2017, 2024 and 2025.

Catenary

When the cable load is its own uniform weight per unit cable length, the curve is a catenary.

$$y = k \cosh(x/k)$$

Here $k=H/w$ under the usual notation. The derivation comes from cable equilibrium and $ds^2=dx^2+dy^2$.

QB: Catenary proof/conditions appear in 2015 and related papers.

Cable Length Approximation

For a shallow parabolic cable with equal-level supports:

$$S \approx L + \frac{8d^2}{3L} - \frac{32d^4}{5L^3}$$

Use only when sag/span is suitably small.

QB: Useful for span/length cable numericals.

Formula Book

Category	Formula	Expression / condition
Centroid	**Composite area**	$\bar{x} = \Sigma A_i x_i / \Sigma A_i$; $\bar{y} = \Sigma A_i y_i / \Sigma A_i$ Cut-out/hole: use negative A.
Centroid	**Triangle**	From base: $\bar{y} = h/3$; from a vertical side for right triangle: $\bar{x} = b/3$ Centroid divides a median 2:1 from vertex.
Centroid	**Semicircle area**	$\bar{y} = 4r/(3\pi)$ from diameter On symmetry axis.
Centroid	**Quarter circle area**	$\bar{x} = \bar{y} = 4r/(3\pi)$ From the two straight sides.
Centroid	**Circular arc ±β**	$\bar{x} = r \sin\beta / \beta$ β in radians.
Centroid	**Circular sector ±β**	$\bar{x} = 2r \sin\beta / (3\beta)$ β in radians.
Area MOI	**Rectangle centroidal**	$I_x = bh^3/12$; $I_y = hb^3/12$ Area units to fourth power.
Area MOI	**Triangle base / centroid**	I_x,base = $bh^3/12$; I_x,G = $bh^3/36$ Centroidal axis parallel to base.
Area MOI	**Circle centroidal diameter**	$I_x = I_y = \pi r^4/4$ Polar $J = \pi r^4/2$.
Area MOI	**Parallel-axis theorem**	$I = \bar{I} + Ad^2$ d perpendicular between parallel axes.
Area MOI	**Polar theorem**	$J = I_x + I_y$ Axes intersect at same point; z perpendicular.
Area MOI	**Radius of gyration**	$k = \sqrt{(I/A)}$ Length unit.
Product Inertia	**Definition**	$P_{xy} = \int xy\, dA$ Can be +, − or 0.
Product Inertia	**Transfer**	$P_{xy} = \bar{P}_{xy} + A\, d_x\, d_y$ Signs of d_x,d_y matter.
Product Inertia	**Principal values**	Imax,min = $(I_x+I_y)/2 \pm \sqrt{\{[(I_x-I_y)/2]^2 + P_{xy}^2\}}$ Principal product inertia becomes zero.
Mass MOI	**Parallel-axis**	$I = \bar{I} + md^2$ Use mass, not area.
Mass MOI	**Slender rod center**	$I = mL^2/12$ Axis perpendicular to rod through center.

Category	Formula	Expression / condition
Mass MOI	**Slender rod end**	$I = mL^2/3$ Axis perpendicular to rod through end.
Mass MOI	**Solid cylinder/disk axis**	$I = mr^2/2$ Geometric longitudinal axis / disk normal axis.
Mass MOI	**Solid sphere diameter**	$I = 2mr^2/5$ Any diameter.
Pappus	**Surface theorem**	$S = 2\pi\bar{y}L$ Curve length L × distance traveled by curve centroid.
Pappus	**Volume theorem**	$V = 2\pi\bar{y}A$ Area A × distance traveled by area centroid.
Statics	**2D equilibrium**	$\Sigma F_x=0$; $\Sigma F_y=0$; $\Sigma M=0$ Use consistent signs.
Statics	**Resultant**	$R = \sqrt{[(\Sigma F_x)^2+(\Sigma F_y)^2]}$ Angle from atan2 concept / quadrant check.
Beam	**UDL resultant**	$W = wL$ at $L/2$ Uniform load.
Beam	**Triangular UVL resultant**	$W = \frac{1}{2}wL$ at $L/3$ from heavy end $2L/3$ from zero end.
Friction	**Limiting friction**	$F = \mu N$ At impending motion only.
Friction	**Angle relation**	$\tan\varphi = \mu$; at repose $\alpha=\varphi$ Static limiting state.
Belt	**Belt friction**	$T_1 = T_2 \exp(\mu\theta)$ θ in radians.
Cable	**Parabolic cable ordinate**	$y = wx^2/(2H)$ w per horizontal length.
Cable	**Horizontal tension**	$H = wL^2/(8d)$ Equal-level supports.
Cable	**Cable tension**	$T = \sqrt{[H^2+(wx)^2]}$ x from lowest point.
Catenary	**Catenary equation**	$y = k \cosh(x/k)$ With chosen origin/constant convention.
Dynamics	**Newton II**	$\Sigma F = ma$ Use mass; in FPS $m=W/g$.
Dynamics	**Constant acceleration**	$v=u+at$; $s=ut+\frac{1}{2}at^2$; $v^2=u^2+2as$ One-dimensional constant a.
Work–Energy	**Work**	

Category	Formula	Expression / condition
		$W = Fs \cos\theta$ Force constant.
Work–Energy	**Work–kinetic energy**	$W_{net} = \Delta KE = ½m(v_2^2 - v_1^2)$ Scalar energy method.
Impulse	**Linear impulse–momentum**	$\int F dt = m\, v_2 - m\, v_1$ Vector relation.
Impulse	**Angular impulse–momentum**	$\int M dt = I(\omega_2 - \omega_1)$ For constant I about chosen axis.
Projectile	**Projectile**	$x = v \cos\theta \cdot t$; $y = v \sin\theta \cdot t - ½gt^2$ No air resistance.

Theory + Proof / Derivations

Centroid of a Triangle: $\bar{y} = h/3$

Very High · 2016, 2018, 2021 note tags · Centroid prepared note pp. 5–8

1. Take a horizontal elemental strip at height y and thickness dy.
2. By similar triangles, strip width = $b(h-y)/h$, so $dA = b(h-y)dy/h$.
3. Total area $A = bh/2$.
4. First moment about base: $\int_0^h y\,dA = \int_0^h y \cdot b(h-y)/h\,dy = bh^2/6$.
5. Therefore $\bar{y} = (bh^2/6)/(bh/2) = h/3$.

> **Result:** $\bar{y} = h/3$ from the base; equivalently $2h/3$ from the opposite vertex.

Perpendicular-Axis Theorem: $J = I_x + I_y$

Must Do · Repeated theory · Moment of Inertia lecture slide

1. For elemental area dA at (x,y), distance from perpendicular z-axis is r.
2. $r^2 = x^2 + y^2$.
3. $J = \int r^2 dA = \int (x^2+y^2) dA$.
4. Split the integral: $J = \int x^2 dA + \int y^2 dA = I_y + I_x$.

> **Result:** $J_O = I_x + I_y$ for mutually perpendicular axes meeting at O.

Parallel-Axis Theorem for Area

Must Do · Repeated theorem/application · MOI slides + class note

1. Let $\bar{x}$-axis pass through the centroid and x-axis be parallel at distance d.
2. For elemental area, $y = \bar{y}' + d$.
3. $I_x = \int (\bar{y}'+d)^2 dA = \int \bar{y}'^2 dA + 2d\int \bar{y}' dA + d^2 \int dA$.
4. Because $\bar{x}$-axis is centroidal, $\int \bar{y}' dA = 0$.
5. Thus $I_x = \bar{I}_x + Ad^2$.

> **Result:** $I = \bar{I} + Ad^2$.

Triangle Area MOI about Centroidal Axis

High · Common derivation · MOI lecture slide / class note

1. Using a horizontal strip, width at y from the base is $b(1-y/h)$.
2. First obtain I about base: $I_{base} = \int_0^h y^2 \cdot b(1-y/h)dy = bh^3/12$.
3. Triangle centroid is $h/3$ from base and $A=bh/2$.
4. By parallel-axis theorem: $I_{base} = I_G + A(h/3)^2$.
5. $I_G = bh^3/12 - (bh/2)(h^2/9) = bh^3/36$.

Result: $I_{x,G} = bh^3/36$.

Pappus–Guldinus Surface Theorem

High · 2015/2018/2023 family · Pappus-Guldinus lecture slides

1. Take an elemental curve length dL at distance y from the axis.
2. On a full revolution it generates $dS = (2\pi y)dL$.
3. Integrate: $S = 2\pi\int y\, dL$.
4. By the centroid definition of a line, $\bar{y} = (\int y\, dL)/L$.
5. Therefore $S = 2\pi\bar{y}L$.

Result: Surface area = generating curve length × distance traveled by its centroid.

Pappus–Guldinus Volume Theorem

High · 2018/2023/2025 family · Pappus-Guldinus lecture slides

1. Take elemental area dA at distance y from the axis.
2. Its revolution sweeps $dV = (2\pi y)dA$.
3. Integrate: $V = 2\pi\int y\, dA$.
4. For area centroid, $\bar{y} = (\int y\, dA)/A$.
5. Hence $V = 2\pi\bar{y}A$.

Result: Volume = generating area × distance traveled by its centroid.

Work–Kinetic Energy Principle

Must Do · Repeated, including 2023 · Work, Power and Energy note

1. Newton's second law along the path: $R = ma$.
2. Use $a = v\,dv/ds$, so $R\,ds = m\,v\,dv$.
3. Integrate from state 1 to 2: $\int R\,ds = \int m\,v\,dv$.
4. Left side is net work; right side is $\frac{1}{2}m(v_2^2 - v_1^2)$.

> **Result:** $W_{net} = KE_2 - KE_1 = \Delta KE$.

Linear Impulse–Momentum Principle

Must Do · Repeated proof · Impulse Momentum note

1. From Newton II: $F = m\,dv/dt$ for constant mass.
2. Multiply by dt: $F dt = m\,dv$.
3. Integrate $t_1 \to t_2$ and $v_1 \to v_2$.
4. $\int F dt = m(v_2 - v_1)$.

> **Result:** Impulse equals change in linear momentum.

Belt Friction: $T_1 = T_2 \exp(\mu\theta)$

Very High · 2015/backlog 2023/2024 family · Friction note + question papers

1. Consider a differential belt element subtending $d\theta$ with tension T and $T+dT$.
2. At impending slip, differential friction $dF = \mu\,dN$.
3. Tangential equilibrium (neglecting second-order small terms) gives $dT = \mu\,dN$.
4. Normal equilibrium gives $dN \approx T\,d\theta$.
5. Therefore $dT/T = \mu\,d\theta$.
6. Integrate from T_2 to T_1 and 0 to θ: $\ln(T_1/T_2) = \mu\theta$.

> **Result:** $T_1/T_2 = \exp(\mu\theta)$, with θ in radians.

Parabolic Cable Equation

Very High · 2017/2024/2025 family · Flexible Cord note

1. Take the lowest point as origin and a cable segment from 0 to x.
2. Horizontal component of tension H is constant.
3. Uniform load over horizontal projection x is wx.
4. Vertical equilibrium gives $T \sin\theta = wx$; horizontal gives $T \cos\theta = H$.
5. Thus $\tan\theta = wx/H = dy/dx$.
6. Integrate with y=0 at x=0: $y = wx^2/(2H)$.
7. At support x=L/2, y=d, so $H=wL^2/(8d)$.

Result: $y=wx^2/(2H)$; $H=wL^2/(8d)$.

Catenary Equation — Core Derivation Path

High · 2015 family · Flexible Cord note

1. For self-weight w per unit cable length, vertical load on a segment is ws where s is arc length from the lowest point.
2. With constant horizontal tension H, $\tan\theta=ws/H$.
3. Since $dy/dx=\tan\theta$ and $ds^2=dx^2+dy^2$, differential equations relate s and x.
4. Let k=H/w. Integration leads to hyperbolic functions.
5. With the conventional origin, the catenary may be written $y=k \cosh(x/k)$, up to a vertical datum constant.

Result: $y = k \cosh(x/k)$ under the convention used in the supplied notes.

Regular Question Bank Analysis

Rank	Topic	Priority	Evidence
1	**Centroid + Composite Area**	Must Do	Centroid definition note-tagged 2015–2022; composite centroid remains present in 2023–2025.
2	**Area Moment of Inertia + Parallel Axis**	Must Do	Appears in essentially every supplied regular-paper era and recent papers.
3	**FBD + Equilibrium + Support Reactions**	Must Do	FBD definition note-tagged 2015,16,17,18,21,22; recent papers continue it.
4	**Shear Force & Bending Moment**	Must Do	Definition note-tagged 2016,17,18,19,21,22; beam numericals recur in recent papers.
5	**Friction / Angle of Repose / Sliding-Tipping**	Must Do	Repeated across historical, recent regular and backlog papers.
6	**Work–Energy + Impulse–Momentum**	Must Do	Core dynamics proof/numerical family throughout the bank.
7	**Pulley**	Very High	Prepared note tags 2017,18,20,22 for a repeated pattern; recent papers also use it.
8	**Truss**	Very High	Truss numericals repeatedly request selected members or all members.
9	**Product of Inertia / Principal MOI**	Very High	Z-section problem note-tagged 2017–2022; 2023/2025 continue principal/product-inertia themes.
10	**Flexible Cord / Cable**	Very High	Catenary/parabolic cable questions recur, especially 2015, 2017, 2024, 2025 families.
11	**Mass MOI**	High	Frustum/disk/cylinder/sphere/mallet patterns across old and recent papers.
12	**Pappus–Guldinus**	High	Surface/volume applications recur in 2015/2018/2023/2025 families.
13	**Wheel least force / obstacle**	High	Prepared note tags 2015,16,17,21 and backlog repeats it.
14	**Wedge**	High	Repeated in 2016 and recent regular/backlog statics.
15	**Projectile / Plane Motion**	High	Motorcycle problem note-tagged 2015,18,21 and appears as a classic pattern.

Year Map

Year	Question block 1	Question block 2
2015	FBD/wheel; cable+beam; friction/belt; truss	Centroid/Pappus; Area MOI; impulse+mass MOI; work-energy/projectile
2016	FBD/resultant; beam+pulley; truss+wheel; friction+wedge	Centroid; Area MOI; Work/CG; Impulse/jet
2017	FBD+wheel; beam+cable; pulley+friction; truss+plane motion	Centroid; MOI; product inertia; impulse
2018	Centroid; Area MOI; Pappus/Product-type; Work+Impulse	FBD/derrick; beam+pulley; truss; friction/plane motion
2019	Centroid; Area/Polar MOI; Product Inertia; Dynamics/Work	Resultant/FBD; beam+pulley; truss; friction/belt/wedge

Year	Question block 1	Question block 2
2020*	Exceptional set arrangement	Exceptional set arrangement
2021	FBD/wheel; cable+beam; friction; impulse/work/projectile	Centroid; Area/Product MOI; truss; impulse
2023	Centroid; Area MOI; Principal/Mass MOI; Dynamics/Work/ Impulse	FBD/supports; beam+pulley; truss; friction/cable/wedge
2024	FBD/supports; beam+cable; truss+pulley; friction+wedge	Centroid; Area/Mass MOI; dynamics/impulse; impact/tipping
2025	FBD/cylinder; beam+cable; truss+pulley; friction+wedge	Centroid; Work+Area MOI; Mass MOI/jet; Product MOI/ dynamics

Complete High-Frequency Question Bank Solution

#1 · Centroid of area bounded by $y^2=4x$ and $x^2=4y$

Pattern: Find the centroid of the first-quadrant area bounded by $y^2=4x$ and $x^2=4y$.

Part A · Centroid · 2022 note-tag, 2024 family · Prepared centroid note + verified calculation

1. Find intersections: substitute $y=x^2/4$ into $y^2=4x$ → $x=0$ or 4, giving (0,0) and (4,4).
2. Use horizontal strips: $x_{right}=2\sqrt{y}$ and $x_{left}=y^2/4$ for $0\le y\le 4$.
3. Area $A=\int_0^4(x_{right}-x_{left})dy=16/3$.
4. First moment about y-axis: $Q_y=\int_0^4 \tfrac{1}{2}(x_{right}^2-x_{left}^2)dy=48/5$.
5. $\bar{x}=Q_y/A=(48/5)/(16/3)=9/5$.
6. The region is symmetric about $y=x$, therefore $\bar{y}=\bar{x}=9/5$.

Final: $(\bar{x},\bar{y})=(1.8\text{ in}, 1.8\text{ in})$
Shortcut: Symmetry catches the second coordinate for free.

#2 · Composite L-section centroid + MOI

Pattern: For a 2×8 in vertical rectangle and 7×2 in top rectangle forming an L-section, find centroid and centroidal I_x, I_y.

Part A · Area MOI · Core template · Shuvo Dip/Hamid class-note pattern + verified arithmetic

1. Areas: $A_1=2\times8=16\text{ in}^2$; $A_2=7\times2=14\text{ in}^2$.
2. Centroids from bottom-left reference: $(x_1,y_1)=(1,4)$, $(x_2,y_2)=(3.5,9)$.
3. $\bar{x}=(16\times1+14\times3.5)/30=2.1667$ in; $\bar{y}=(16\times4+14\times9)/30=6.3333$ in.
4. Use $I=\bar{I}+Ad^2$ for each rectangle about centroidal x-axis and sum: $I_x=276.67\text{ in}^4$.
5. Repeat about centroidal y-axis: $I_y=109.17\text{ in}^4$.

Final: $\bar{x}=2.167$ in, $\bar{y}=6.333$ in; $I_x\approx276.67\text{ in}^4$; $I_y\approx109.17\text{ in}^4$
Shortcut: Coordinates first; transfer distances are part-centroid minus composite-centroid.

#3 · Z-section product of inertia + principal MOI

Pattern: For the supplied 8×5×1 in Z-section, find Pxy, maximum/minimum MOI and radii of gyration.

Part A · Product Inertia · 2017, 2018, 2019, 2020*, 2021, 2022 note-tag · 3.1 Product of Inertia prepared note

1. Split the section into three rectangles about centroidal x′ and y′ axes.
2. For each rectangle use $P_{xy}=\bar{P}+A\,dx\,dy$. Each rectangle has $\bar{P}=0$ about its own centroidal axes.
3. Top flange: +35 in^4; web: 0; bottom flange: +35 in^4. Hence $P_{xy}=70$ in^4.
4. From the supplied solution, $I_x=141.34$ in^4 and $I_y=61.33$ in^4.
5. $I_{avg}=(141.34+61.33)/2=101.335$ in^4.
6. $R=\sqrt{[(141.34-61.33)/2]^2+70^2}=80.625$ in^4.
7. $I_{max}=181.96$ in^4; $I_{min}=20.71$ in^4.
8. Total area $A=16$ in^2, so $k_{max}=\sqrt{(181.96/16)}=3.372$ in and $k_{min}\approx\sqrt{(20.71/16)}=1.138$ in.

> **Final:** $P_{xy}=70$ in^4; $I_{max}=181.96$ in^4; $I_{min}=20.71$ in^4; $k_{max}=3.372$ in; $k_{min}\approx1.138$ in
> **Shortcut:** For product inertia, dx·dy sign comes from the quadrant.

#4 · Mallet — mass MOI and radius of gyration

Pattern: 3 ft handle weighs 3.14 lb; cylindrical head weighs 16.16 lb. Find radius of gyration about y-axis.

Part A · Mass MOI · 2022 note-tag, 2025 family · 3.2 Moment of Inertia of Masses prepared note

1. Convert each weight to mass using $m=W/g$, $g=32.2$ ft/s^2.
2. Handle is a slender rod: $I_{y1}=\bar{I}_y+m_1d_1^2=m_1L^2/12+m_1(1\text{ ft})^2\approx0.17$ $slug\cdot ft^2$.
3. Head is a cylinder. Use its centroidal MOI about a transverse axis and shift by $d_2=2.75$ ft as in the supplied geometry.
4. Supplied calculation gives $I_{y2}\approx3.83$ $slug\cdot ft^2$.
5. Total $I_y\approx4.00$ $slug\cdot ft^2$.
6. Total mass=(3.14+16.16)/32.2 slug.
7. $k=\sqrt{(I/m)}\approx2.59$ ft.

> **Final:** $I_y\approx4.00$ $slug\cdot ft^2$; $k_y\approx2.59$ ft
> **Shortcut:** In FPS: never use lb directly as mass; divide weight by g.

#5 · Inclined block — resultant and net work

Pattern: $W=200$ lb, $\alpha=35°$, $Q=125$ lb at $\beta=25°$ above the incline, $\mu=1/3$, displacement=12 ft up the incline. Find resultant along motion and net work.

Part A · Work–Energy · Backlog 2023 family · Backlog 2023 + verified calculation

1. Normal equilibrium: $N=W\cos35°-Q\sin25°=111.00$ lb.
2. Friction opposing upward motion: $F_f=\mu N=(1/3)(111.00)=37.00$ lb.
3. Along incline (up positive): $R=Q\cos25°-W\sin35°-F_f$.
4. $R=-38.43$ lb, so the resultant acts down the incline.
5. Net work over 12 ft: $W_{net}=R\cdot s=-38.43\times12=-461.13$ ft·lb.
6. Since $W_{net}<0$, kinetic energy/velocity decreases.

Final: Resultant = 38.43 lb down the incline; $W_{net}\approx-461.13$ ft·lb
Shortcut: Normal force changes because Q has a component away from the plane.

#6 · Motorcycle projectile — maximum landing height

Pattern: A motorcycle leaves at 70 mph, 30°, crosses a 20 ft ditch. Launch point is 10 ft above ground. Find maximum landing height h.

Part A · Dynamics · 2015, 2018, 2021 note tags · 4.1 Dynamics prepared note

1. Convert 70 mph≈102.67 ft/s (or use SI consistently).
2. Horizontal travel $x=20$ ft gives $t=x/(v\cos30°)$.
3. Vertical rise relative to launch: $y=v\sin30°\cdot t-\frac{1}{2}gt^2\approx10.7$ ft (minor difference depends on rounded conversions).
4. Add the 10 ft launch elevation.
5. The supplied note calculation gives $h\approx20.73$ ft.

Final: $h_{max}\approx20.73$ ft (using the supplied-note conversion/rounding)
Shortcut: Do not use the range formula because launch and landing elevations differ.

#7 · Wheel over block — least force

Pattern: For $W=1000$ lb, radius 3 ft and the supplied block geometry, find the least force F and its direction.

Part B · Equilibrium · 2015, 2016, 2017, 2021, Backlog 2023 · Equilibrium prepared note

1. At impending rotation over the corner, ground reaction becomes zero and the wheel pivots about the block corner.
2. Geometry gives $\alpha'=\sin^{-1}(2/3)=41.81°$ and $\alpha=\alpha'+15°=56.81°$.
3. Take moments about the corner: $F\sin(\theta+56.81°)=W\cos 56.81°$.
4. Thus $F=547.42/\sin(\theta+56.81°)$.
5. For minimum F, denominator must be 1: $\theta+56.81°=90°$.
6. Therefore $\theta=33.19°$ and $F_{min}=547.42$ lb.

Final: $F_{min}=547.42$ lb at $\theta=33.19°$
Shortcut: Least-force problems: optimize the trigonometric denominator after the moment equation.

#8 · Beam SFBM — complete cut solution

Pattern: For the prepared beam with UDL, 30-kip point load, 20-kip-ft couple and triangular load, find reactions and V/M at section (i).

Part B · SFBM · 2018, 2019, 2020*, 2021 · 2.1 SFBM prepared note

1. Replace the UDL and triangular UVL by their point-resultants at their centroids.
2. From whole-beam moment equilibrium about B, the note obtains $V_E=36$ kip.
3. From vertical equilibrium, $V_B=109$ kip. Horizontal reaction is zero.
4. Cut at section (i). Left-side vertical equilibrium gives $V=-21$ kip.
5. Left-side moment equilibrium about the cut gives $M=-48$ kip·ft.
6. Checking from the right side gives the same V and M, confirming signs and lever arms.

Final: $V_B=109$ kip; $V_E=36$ kip; $V(i)=-21$ kip; $M(i)=-48$ kip·ft
Shortcut: A pure couple enters the BM equation but not the shear equation.

#9 · Pulley + beam equilibrium — W=21,500 lb

Pattern: Ideal pulley system supports W and pulls a beam carrying 700 lb plus 2 kip/ft UDL. Find W.

Part B · Pulley · 2016 family · 2.2 Pulley prepared note

1. Count supporting rope segments on the movable pulley block. The beam-end rope tension is W/4 for this routing.
2. The UDL is 2,000 lb/ft over 6 ft, resultant 12,000 lb acting 3 ft from its loaded-span center reference shown in the note.
3. Take moments about beam support A exactly as in the supplied geometry:
4. $-(2000\times6\times3) - (700\times10) + (W/4\times8)=0$.
5. Solve W=21,500 lb.

> **Final:** W=21,500 lb
> **Shortcut:** Do not reuse W/4 for a different pulley diagram; derive it from the rope path.

#10 · Truss — section method template

Pattern: For the supplied sloping truss, find selected member forces using a section.

Part B · Truss · 2017, 2022 note-tag, Recent family · 3.1 Truss prepared note

1. Resolve the inclined external loads and find whole-truss reactions first.
2. The prepared solution obtains $G_y=10.7648$ k, $A_x=5.5$ k and $A_y=-2.4347$ k.
3. Pass section (i)-(i) through the target members.
4. Take moment about C so two cut-member forces vanish from the equation.
5. Solving gives $F_{GJ}=-9.9278$ k relative to assumed tension.
6. Negative assumed tension means compression.

> **Final:** $F_{GJ}\approx9.928$ kips (Compression) for the supplied example
> **Shortcut:** Choose a moment center that eliminates the maximum number of unknown cut forces.

#11 · Belt friction derivation

Pattern: Prove $T_1=T_2\exp(\mu\theta)$.

Part B · Friction · 2015, Backlog 2023, 2024 · Friction notes + QB

1. Take a small belt element subtending $d\theta$.
2. Let tensions be T and $T+dT$ and normal reaction be dN.
3. At impending slip, $dF=\mu dN$.
4. Tangential equilibrium gives $dT\approx dF=\mu dN$.
5. Normal equilibrium for a small element gives $dN\approx T\,d\theta$.
6. Hence $dT/T=\mu d\theta$.
7. Integrate: $\ln(T_1/T_2)=\mu\theta$, so $T_1=T_2\exp(\mu\theta)$.

Final: $T_1/T_2=\exp(\mu\theta)$, θ in radians
Shortcut: The exponential formula is invalid if θ is substituted in degrees.

#12 · Parabolic cable — sag/tension equations

Pattern: Derive the parabolic cable equation and support tension relation.

Part B · Flexible Cord · 2017, 2024, 2025 · Flexible Cord prepared note

1. Take the lowest point O as origin and a cable segment ending at (x,y).
2. Horizontal tension component is H; load over x is wx.
3. Equilibrium: $T\cos\theta=H$ and $T\sin\theta=wx$.
4. Thus $dy/dx=\tan\theta=wx/H$.
5. Integrate: $y=wx^2/(2H)$.
6. At $x=L/2$, $y=d$ gives $H=wL^2/(8d)$.
7. At any x, $T=\sqrt{H^2+(wx)^2}$; maximum at the supports.

Final: $y=wx^2/(2H)$; $H=wL^2/(8d)$; $T_{max}=\sqrt{[H^2+(wL/2)^2]}$
Shortcut: w is load per horizontal length for a parabolic cable.

#13 · Linear impulse–momentum proof

Pattern: State and prove the principle of linear impulse and momentum.

Part A · Impulse · 2016, 2018, 2023, 2024 family · Impulse Momentum note

1. State: total impulse of external forces over a time interval equals the change in linear momentum.
2. Start with $F=ma=m\,dv/dt$.
3. Rearrange $Fdt=m\,dv$.
4. Integrate from t1 to t2: $\int Fdt=m\int dv$.
5. Therefore $\int Fdt=m(v2-v1)$. For constant resultant R, $R\Delta t=\Delta(mv)$.

Final: $\int Fdt = mv_2-mv_1$
Shortcut: Write it as a vector principle unless the problem is one-dimensional.

#14 · Pappus cone surface / paint template

Pattern: Use Pappus-Guldinus to find the lateral area of a cone of radius r and height h, then paint for inside and outside.

Part A · Pappus · 2015, Backlog 2023 family · Pappus lecture slide + QB

1. The generating line is the slant length $l=\sqrt{r^2+h^2}$.
2. Centroid of a straight generating segment lies at its midpoint, distance r/2 from the axis.
3. Pappus surface theorem: $Sone=2\pi(r/2)l=\pi rl$.
4. For both inside and outside surfaces of a thin shell, $Stotal=2\pi rl$.
5. Paint gallons=Stotal/(coverage per gallon), if both surfaces are intended and no base/top areas are specified.

Final: One lateral surface = $\pi r\sqrt{r^2+h^2}$; both faces = $2\pi r\sqrt{r^2+h^2}$
Shortcut: State clearly which surfaces are included; question wording may exclude base/opening.

#15 · Force–mass acceleration block

Pattern: 300 N block on horizontal plane, P=200 N applied at 30° downward, μk=0.2. From rest, find position and velocity at 5 s.

Part A · Dynamics · Prepared practice · 4.1 Dynamics note

1. Vertical component of P is 200sin30°=100 N downward, so N=300+100=400 N.
2. Friction=μN=0.2×400=80 N.
3. Horizontal net force=200cos30°−80.
4. Mass=300/9.8 kg. Therefore a≈3.04 m/s².
5. From rest: s=½at²=½×3.04×25=38.05 m.
6. v=at=3.04×5=15.2 m/s.

Final: a≈3.04 m/s²; s≈38.05 m; v≈15.2 m/s
Shortcut: A downward angled pull/push increases N; an upward component decreases N.

#16 · FBD + transmissibility + two-force member — exam answer

Pattern: Define FBD, transmissibility of force, two-force member and equilibrium; state how to draw the FBD.

Part B · Equilibrium · 2015, 2016, 2017, 2018, 2021, 2022 note-tag, Backlog 2023 · Equilibrium prepared note

1. Free-body diagram: isolate the body/member and show every external force, applied load, weight, support reaction and friction acting on it.
2. Transmissibility: for a rigid body, a force may be moved anywhere along the same line of action without changing the external equilibrium/motion, provided magnitude and direction are unchanged.
3. Two-force member: if only two forces act on an equilibrium member, the forces must be equal, opposite and collinear.
4. Equilibrium in a coplanar system requires $\Sigma F_x=0$, $\Sigma F_y=0$ and $\Sigma M=0$.
5. Drawing shortcut: isolate tension/compression members first; then replace each support by its correct reaction components; add self-weight unless the problem says weightless.

Final: Write definitions + a clean isolated sketch; show support reactions and all external actions.
Shortcut: বাংলা মনে রাখুন: “আলাদা করো → সব বাহ্যিক বল দেখাও → সাপোর্টকে reaction দিয়ে বদলাও।”

#17 · Coplanar resultant — component method

Pattern: For the prepared concurrent-force example, determine resultant magnitude and direction.

Part B · Force System · 2016, 2019 family · Equilibrium prepared note

1. Choose +x to the right and +y upward. Resolve every inclined force before adding.
2. The prepared example gives $\Sigma F_x=37.79$ kN and $\Sigma F_y=-21.81$ kN.
3. Magnitude: $R=\sqrt{[(\Sigma F_x)^2+(\Sigma F_y)^2]}=43.633$ kN.
4. Reference angle: $\tan^{-1}(|\Sigma F_y|/|\Sigma F_x|)=29.99°$.
5. Because $\Sigma F_x>0$ and $\Sigma F_y<0$, the resultant lies in quadrant IV: 29.99° below the +x-axis.

Final: R≈43.633 kN, 29.99° below +x-axis for the prepared example
Shortcut: Never decide the quadrant from $\tan^{-1}$ alone; use the signs of ΣF_x and ΣF_y.

#18 · Triangle centroid — full derivation

Pattern: Derive the centroid of a triangle of base b and height h.

Part A · Centroid · 2016, 2018, 2021 note-tags · Centroid prepared note

1. Take a horizontal strip at height y with thickness dy. From similar triangles, strip width = $b(h-y)/h$.
2. Thus $dA=[b(h-y)/h]dy$ and $A=bh/2$.
3. First moment about the base: $\int y\, dA = \int_0^h y\cdot b(h-y)/h\, dy = bh^2/6$.
4. Therefore $\bar{y}=(\int y\, dA)/A=(bh^2/6)/(bh/2)=h/3$ from the base.
5. Similarly, for the usual right-triangle reference, $\bar{x}=b/3$ from the perpendicular side; in any triangle the centroid is the intersection of the medians and divides each median 2:1.

Final: $\bar{y}=h/3$ from the base; centroid lies at the intersection of medians.
Shortcut: মনে রাখুন: triangle centroid = base থেকে h/3, vertex থেকে 2h/3.

#19 · Perpendicular-axis theorem — $J = I_x + I_y$

Pattern: Prove the perpendicular-axis theorem for a plane area.

Part A · Area MOI · 2018, 2019, 2021, Backlog 2023 · MOI lecture slide + prepared notes

1. For an element dA at coordinates (x,y), its distance from the z-axis normal to the plane is r, where $r^2=x^2+y^2$.
2. Polar moment: $J_o=\int r^2 dA$.
3. Substitute $r^2=x^2+y^2$: $J_o=\int x^2 dA+\int y^2 dA$.
4. By definition, $\int x^2 dA=I_y$ and $\int y^2 dA=I_x$.
5. Hence $J_o=I_x+I_y$, provided x and y lie in the plane and all three axes intersect at the same point.

Final: $J_o = I_x + I_y$
Shortcut: Condition is essential: mutually perpendicular axes through one common point.

#20 · Parallel-axis theorem — $I = \bar{I} + Ad^2$

Pattern: Prove the transfer/parallel-axis formula for an area.

Part A · Area MOI · 2015 family, 2018+ · MOI slide + prepared notes

1. Let $\bar{x}$ be a centroidal axis and x a parallel axis at distance d.
2. For an element dA, write its distance from x as $y=y'+d$.
3. $I_x=\int(y'+d)^2 dA = \int y'^2 dA + 2d\int y' dA + d^2\int dA$.
4. Because $\bar{x}$ passes through the centroid, $\int y' dA=0$.
5. Therefore $I_x=\bar{I}_x+Ad^2$.

Final: $I = \bar{I} + Ad^2$ (area); for mass MOI, $I = \bar{I} + md^2$.
Shortcut: The transfer distance d is always the perpendicular distance between parallel axes.

#21 · Angle of friction vs angle of repose

Pattern: Distinguish between angle of friction and angle of repose.

Part B · Friction · 2015, 2016, 2019, 2024, Backlog 2023 · Friction prepared note + QB

1. Angle of friction φ: at limiting friction, the resultant reaction R is inclined to the normal N by φ.
2. Since $\tan\varphi=F_{lim}/N$ and $F_{lim}=\mu N$, $\tan\varphi=\mu$.
3. Angle of repose α: the inclination of a rough plane to the horizontal at which a body is just about to slide under its own weight.
4. At impending slide on the plane, $W\sin\alpha=\mu W\cos\alpha$, so $\tan\alpha=\mu$.
5. Therefore $\alpha=\varphi$ for the same pair of contact surfaces under limiting equilibrium.

> **Final:** $\tan\varphi=\mu$, $\tan\alpha=\mu \Rightarrow \alpha=\varphi$ at limiting equilibrium.
> **Shortcut:** Friction angle belongs to the reaction triangle; repose angle belongs to the inclined plane.

#22 · Catenary — governing equation

Pattern: Show that a cable carrying only its own uniformly distributed weight forms a catenary.

Part B · Flexible Cord · 2015, 2021 family · Flexible Cord note + regular QB

1. For a cable whose load is uniform per unit cable length, use the lowest point as origin and let H be the constant horizontal component of tension.
2. For a cable element, vertical equilibrium relates the vertical tension component to the weight of cable length s.
3. With $\tan\theta=dy/dx$ and $ds^2=dx^2+dy^2$, the equilibrium differential equation integrates to the hyperbolic form.
4. Writing $k=H/w$, the curve is $y=k\cosh(x/k)$ when the origin convention is chosen as in the course question; an equivalent vertically shifted form is $y=k[\cosh(x/k)-1]$ when $y=0$ is set at the lowest point.
5. State the coordinate convention in the exam before giving the final expression.

> **Final:** Catenary form: $y = k\cosh(x/k)$, $k=H/w$ (or the vertically shifted equivalent).
> **Shortcut:** Parabola → load uniform per horizontal span; catenary → self-weight uniform per cable length.

#23 · Cone with cylindrical hole — center of gravity

Pattern: Solid cone D=20 in, h=30 in has a coaxial cylindrical hole d=8 in, depth=6 in from the base. Locate C.G.

Part A · Centroid / C.G. · 2016 · 2016 regular QB; derived from standard composite-volume centroid method

1. Use the base as reference. Treat the removed cylinder as negative volume; density cancels because the material is homogeneous.
2. Cone: R=10 in, $V_1=(1/3)\pi R^2h=1000\pi$ in³; its centroid is h/4=7.5 in from the base.
3. Hole: r=4 in, depth=6 in, $V_2=\pi r^2(6)=96\pi$ in³; its centroid is 3 in from the base.
4. Remaining centroid: $\bar{y}=(V_1y_1-V_2y_2)/(V_1-V_2)$.
5. $\bar{y}=[1000\pi(7.5)-96\pi(3)]/[904\pi]\approx7.978$ in from the base.

Final: C.G. ≈ 7.98 in from the base along the geometric axis.
Shortcut: A hole is a negative area/volume/mass in centroid equations.

#24 · Masonry chimney — work against gravity

Pattern: Chimney h=450 ft, OD=22 ft, ID=14 ft, masonry weight density=100 lb/ft³. Find work against gravity during construction.

Part A · Work–Energy · 2016 family, 2023 · 2023 regular QB; derived by integration

1. Cross-sectional masonry area $A=\pi/4(D^2-d^2)=\pi/4(22^2-14^2)=72\pi$ ft².
2. A horizontal slice of thickness dy weighs $dW=\gamma A\,dy$.
3. To place that slice at elevation y requires $dU=y\cdot dW=\gamma Ay\,dy$.
4. Integrate from 0 to h: $U=\gamma A\int_0^h y\,dy=\gamma Ah^2/2$.
5. $U=100(72\pi)(450^2)/2\approx2.290\times10^9$ ft·lb.

Final: Work ≈ 2.290×10^9 ft·lb (≈1.157×10^3 hp·h).
Shortcut: Do not multiply total weight by full height; the average lift height is h/2.

#25 · Jet/steam on fixed blade — impulse–momentum

Pattern: Steam flow rate $\dot{W}$=1.5 lb/s enters a fixed frictionless blade at 500 ft/s and turns through 135°. Find force components on the blade.

Part A · Impulse Momentum · 2015 family, 2024 family · Impulse Momentum prepared note; values checked from the supplied example

1. Mass-flow rate in FPS units: $\dot{m}=\dot{W}/g=1.5/32.2$ slug/s.
2. Take inlet velocity $V_1=(500,0)$ ft/s. A 135° turn gives outlet $V_2=(-500\cos45°,-500\sin45°)$ ft/s.
3. Force on fluid: $F=\dot{m}(V_2-V_1)$. Thus $F_x\approx-39.76$ lb and $F_y\approx-16.47$ lb.
4. The force exerted by the fluid on the fixed blade is equal and opposite.
5. Therefore the blade receives ≈39.76 lb to the right and ≈16.47 lb upward.

Final: Force on blade ≈ (39.76 lb right, 16.47 lb up); resultant ≈43.03 lb.
Shortcut: Always distinguish “force on fluid” from “force on blade”; they are opposite.

Backlog 2023 Dedicated Preparation

Q	Topics	Preparation action
Q1	FBD definitions + member FBD; wheel over block least force	Master B1 definitions and the 547.42-lb least-force template.
Q2	Beam reaction/SF/BM; pulley equilibrium	Do B2 beam workflow plus all 3 prepared pulley routings.
Q3	Belt friction proof; inverted Howe truss	Memorize the derivation logic, not only formula; practice method of sections.
Q4	Sliding vs tipping; wedge; angle friction vs repose	Practice friction-direction logic and separate wedge/block FBDs.
Q5	Centroid/C.G. theory; sector centroid; composite first moment	A1 is a scoring set if the area table is clean.
Q6	Area MOI applications; J=Ix+Iy; polar MOI and radius of gyration	A2 formulas + parallel-axis method.
Q7	Impulse-momentum proof; inclined block work	A4 proof + work example included in solutions.
Q8	Pappus-Guldinus surface/volume; angle friction/repose; horizontal impulse	Revise Pappus theorem statements and impulse sign conventions.

7-Day Plan

1. Day 1 — B1 FBD + equilibrium + wheel. Write definitions from memory twice.
2. Day 2 — B2 Beam + pulley. Solve one beam from both left and right cuts.
3. Day 3 — B3 Truss + B4 belt friction/wedge.
4. Day 4 — A1 Centroid: theory, triangle/sector, composite area.
5. Day 5 — A2 MOI: J theorem, parallel axis, composite MOI.
6. Day 6 — A4 Work–Energy + Impulse proof and numericals; A3 Pappus quick revision.
7. Day 7 — Full 3-hour mock: choose three questions from each printed section; mark units, diagrams and final answers.

Answer-writing rules

- Start every numerical with a clean sketch/FBD and known data.
- Box the governing equation before substitution.
- Carry units on every final quantity.
- If a sign is negative, explain the physical direction/nature instead of deleting the sign.
- For derivations, write assumptions/conditions before equations.
- Do not mix regular-priority statistics with backlog-only prediction.